Supplemental Material
Predicting First-Passage Dynamics in Disordered Systems Exactly: Application to Sparse Networks

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(Dated: August 5, 2025)

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I OCCUPATION PROBABILITY OF THE K-NEIGHBOUR WALK

The general dynamics on the K-neighbour ring lattice may be found by considering first the unbounded dynamics along lattice site n given by via the Master equation

$$P(n, t+1) = \frac{q}{K} \sum_{r=1}^{k} \Big[P(n-r, t) + P(n+r, t) \Big] + (1-q) P(n, t), \tag{S1}$$

where Eq. (1) of the main text is obtained by taking $q = 1$. Equation (S1) may be solved by performing the Fourier transform, $\widehat{f}(\xi) = \sum_{n=-\infty}^{\infty} f(n) \mathrm{e}^{i\xi n}$,

$$\widehat{P}^{(k)}(\xi, t+1) = \left[1 - q + \frac{q}{K} \sum_{r=1}^{k} \cos(r\xi) \right] \widehat{P}^{(k)}(\xi, t), \tag{S2}$$

where we use the superscript k to indicate the number of left or right neighbours ($K = 2k$). After z-transforming the time variable, $\widetilde{f}(z) = \sum_{t=0}^{\infty} z^t f(t)$, and by taking the inverse Fourier transform, we obtain the Lattice Green's

Function, or propagator,

$$\widetilde{P}_{n_0}^{(k)}(n,z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{e^{-i\xi(n-n_0)}}{1-z\left[1-q+\frac{q}{k}\sum_{r=1}^{k}\cos(r\xi)\right]}\mathrm{d}\xi, \tag{S3}$$

where n_0 is the walker's initial site ($\widehat{P}^{(k)}(\xi,0) = e^{i\xi n_0}$).

To simplify Eq. (S3) we use the identity $\sum_{r=1}^{k}\cos(r\xi) = \sin\left(\frac{k\xi}{2}\right)\cos\left(\frac{(k+1)\xi}{2}\right)\csc\left(\frac{\xi}{2}\right)$ to obtain

$$\widetilde{P}_{n_0}^{(k)}(n,z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{e^{-i\xi(n-n_0)}}{1-z\left[1-q+\frac{q}{k}\sin\left(\frac{k\xi}{2}\right)\cos\left(\frac{(k+1)\xi}{2}\right)\csc\left(\frac{\xi}{2}\right)\right]}\mathrm{d}\xi. \tag{S4}$$

To bound the dynamics to a ring lattice we apply the method of images $\widetilde{Q}_{n_0}^{(k)}(n,z) = \sum_{\ell=\infty}^{\infty}\widetilde{P}_{n_0+\ell N}^{(k)}(n,z)$ [1–3], where N is the number of sites in the finite lattice, leading to

$$\widetilde{Q}_{n_0}^{(k)}(n,z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{\sum_{\ell=-\infty}^{\infty} e^{-i\xi(n-n_0-\ell N)}}{1-z\left[1-q+\frac{q}{k}\sin\left(\frac{k\xi}{2}\right)\cos\left(\frac{(k+1)\xi}{2}\right)\csc\left(\frac{\xi}{2}\right)\right]}\mathrm{d}\xi. \tag{S5}$$

To solve the integral in Eq. (S5) we employ the identity [1, 4]

$$\sum_{\ell=-\infty}^{\infty} e^{-i\ell N\xi} = \frac{2\pi}{N}\sum_{\ell=-\infty}^{\infty}\delta\left(\xi-\frac{2\pi\ell}{N}\right), \tag{S6}$$

to perform the integration and find (see ref. [1] for a detailed discussion of integrals of this type)

$$\widetilde{Q}_{n_0}^{(k)}(n,z) = \frac{1}{N(1-z)} + \frac{1}{N}\sum_{\ell=1}^{N-1}\frac{\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)}{1-\frac{z}{k}\left[1-q+\frac{q}{k}\sin\left(\frac{k\ell\pi}{N}\right)\cos\left(\frac{(k+1)\ell\pi}{N}\right)\csc\left(\frac{\ell\pi}{N}\right)\right]}. \tag{S7}$$

Finally, noting that $\lim_{\ell\to 0}\sin\left(\frac{k\ell\pi}{N}\right)\cos\left(\frac{(k+1)\ell\pi}{N}\right)\csc\left(\frac{\ell\pi}{N}\right) = 1$ and also taking $q = 1$, gives Eq. (2) of the main text.

If required, the time dependent dynamics are easily extracted from Eq. (S7) via the inverse z-transform, $f(t) = (2\pi i)\oint \widetilde{f}(z)z^{-t-1}\mathrm{d}z$ with $|z| < 1$ and the integration contour being counterclockwise. Performing this integral leads to

$$Q_{n_0}^{(k)}(n,t) = \frac{1}{N}\sum_{\ell=0}^{N-1}\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)\left[1-q+\frac{q}{k}\sin\left(\frac{k\ell\pi}{N}\right)\cos\left(\frac{(k+1)\ell\pi}{N}\right)\csc\left(\frac{\ell\pi}{N}\right)\right]^t. \tag{S8}$$

The steady state, found from Eq. (S7) via $Q^{(k)}(n,t\to\infty) = \lim_{z\to 1}(1-z)Q^{(k)}(n,z)$, is as expected $Q^{(k)}(n,t\to\infty) = 1/N$ for all n and is independent of k.

I.A Mean squared displacement for the unbounded K-neighbour walk

To understand the effects on the transport process the higher the value of k, we derive the time dependence of the mean square displacement (MSD), $\Delta n(t) = \langle n^2\rangle(t) - \langle n\rangle^2(t)$, in unbounded space. Solving Eq. (S2) we find

$$\widehat{P}_{n_0}^{(k)}(\xi,t) = \left[1-q+\frac{q}{k}\sin\left(\frac{k\xi}{2}\right)\cos\left(\frac{(k+1)\xi}{2}\right)\csc\left(\frac{\xi}{2}\right)\right]^t e^{i\xi n_0}, \tag{S9}$$

where we have taken $P^{(k)}(n,0) = \delta_{n,n_0}$ with $\delta_{a,b}$ the Kronecker delta. Using the fact that $\langle n^2\rangle(t) = -\left.\frac{\partial^2\widehat{P}_{n_0}^{(k)}(\xi,t)}{\partial\xi^2}\right|_{\xi=0}$ and $\langle n\rangle(t) = -i\left.\frac{\partial\widehat{P}_{n_0}^{(k)}(\xi,t)}{\partial\xi}\right|_{\xi=0}$, the MSD in the unbounded lattice,

$$\Delta n(t) = \frac{q}{6}(1+k)(1+K)\,t, \tag{S10}$$

shows that the longer the range of connections the faster the diffusive spread with a coefficient proportional to $1+3k+2k^2$.

I.B Representing the propagator on a finite ring via Chebyshev polynomials

As mentioned in the main text, for $k \in \{1,2,3\}$ one may bypass the need to perform the finite sum in Eq. (S7). Below we show the derivations for each of the case.

I.B.1 The nearest-neighbour ($k=1$) walk

This case has been derived in ref. [3], and we thus simply state the result here:

$$\widetilde{Q}^{(1)}_{n_0}(n,z) = \frac{\cosh\left[\left(N-|n-n_0|\right)\gamma\right] + \cosh\left[|n-n_0|\gamma\right]}{zq\sinh(\gamma)\sinh(N\gamma)} \tag{S11}$$

where $\cosh(\gamma) = 1 + \frac{1}{q}\left[\frac{1}{z}-1\right]$ and $\sinh(\gamma) = \frac{\sqrt{(1-z)\left[1-(1-2q)z\right]}}{qz}$.

I.B.2 The next-nearest-neighbour ($k=2$) walk

When $k=2$ we rewrite the unbounded propagator in Eq. S3 as

$$\begin{aligned}\widetilde{P}^{(2)}_{n_0}(n,z) &= \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{e^{-i\xi(n-n_0)}}{1-z\left[1-\frac{3q}{2}+q\cos^2(\xi)+\frac{q}{2}\cos(\xi)\right]}\mathrm{d}\xi,\\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{e^{-i\xi(n-n_0)}}{2zq\sqrt{\frac{25}{16}+\frac{1}{q}\left[\frac{1}{z}-1\right]}}\left\{\frac{1}{\beta_z^+}\frac{1}{1+\frac{\cos(\xi)}{\beta_z^+}}+\frac{1}{\beta_z^-}\frac{1}{1-\frac{\cos(\xi)}{\beta_z^-}}\right\}\mathrm{d}\xi,\end{aligned} \tag{S12}$$

where $|\beta_z^\pm|>1$ defined by

$$\beta_z^\pm = \sqrt{\frac{25}{16}+\frac{1}{q}\left[\frac{1}{z}-1\right]} \pm \frac{1}{4}, \tag{S13}$$

and where the second line in (S12) has been obtained by finding the zeros of the quadratic polynomial in $\cos(\xi)$ and making a partial fraction expansion. This expansion enables us to compute the integral to obtain

$$\widetilde{P}^{(2)}_{n_0}(n,z) = \frac{1}{2zq\left(\beta_z^+-\frac{1}{4}\right)}\left\{\frac{1}{\sqrt{(\beta_z^+)^2-1}}\frac{(-1)^{n-n_0}}{\left(\sqrt{(\beta_z^+)^2-1}+\beta_z^+\right)^{|n-n_0|}}+\frac{1}{\sqrt{(\beta_z^-)^2-1}}\frac{1}{\left(\sqrt{(\beta_z^-)^2-1}+\beta_z^-\right)^{|n-n_0|}}\right\}. \tag{S14}$$

One may verify normalisation via $\sum_{n=-\infty}^{+\infty}\widetilde{Q}^{(2)}_{n_0}(n,z) = \frac{1}{1-z}$ and obtain the initial condition, i.e. by taking the $z\to 0$ limit, to verify that $\widetilde{Q}^{(2)}_{n_0}(n,0) = \delta_{n.n_0}$. One may also identify how the probability decays to zero al long times by looking at the $z\to 1$ limit, which gives $\widetilde{Q}^{(2)}_{n_0} \simeq \frac{1}{\sqrt{5q}}\frac{1}{\sqrt{1-z}}$, and corresponds to $Q^{(2)}_{n_0}(n,t) \simeq \frac{1}{\sqrt{5\pi q\, t}}$ for $t\to\infty$.

To bound the domain we again employ the method of images shown in Eq. (S5), using Eq. (S6), to deduce

$$\begin{aligned}\widetilde{Q}^{(2)}_{n_0}(n,z) &= \frac{1}{2zq\sqrt{\frac{25}{16}+\frac{1}{q}\left[\frac{1}{z}-1\right]}N}\left\{\sum_{\ell=0}^{N-1}\frac{\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)}{\beta_z^- - \cos\left(\frac{2\pi\ell}{N}\right)}+\sum_{\ell=0}^{N-1}\frac{\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)}{\beta_z^+ + \cos\left(\frac{2\pi\ell}{N}\right)}\right\}\\ &= \frac{1}{N(1-z)}+\frac{1}{2zq\sqrt{\frac{25}{16}+\frac{1}{q}\left[\frac{1}{z}-1\right]}N}\left\{\sum_{\ell=1}^{N-1}\frac{\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)}{\beta_z^- - \cos\left(\frac{2\pi\ell}{N}\right)}-\sum_{\ell=1}^{N-1}\frac{\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)}{-\beta_z^+ - \cos\left(\frac{2\pi\ell}{N}\right)}\right\}\\ &= \frac{1}{2zq\sqrt{\frac{25}{16}+\frac{1}{q}\left[\frac{1}{z}-1\right]}}\left\{\frac{T_{N-|n-n_0|}(\beta_z^-)+T_{|n-n_0|}(\beta_z^-)}{\left[(\beta_z^-)^2-1\right]U_{N-1}(\beta_z^-)}+(-1)^{n-n_0}\frac{T_{N-|n-n_0|}(\beta_z^+)+(-1)^N T_{|n-n_0|}(\beta_z^+)}{\left[(\beta_z^+)^2-1\right]U_{N-1}(\beta_z^+)}\right\},\end{aligned} \tag{S15}$$

where in the last equation trigonometric identities from ref. [3] have been used with $T_r(x)$ and $U_r(x)$ indicating Chebyshev polynomials, respectively, of the first and second kind, of degree r and argument x.

I.B.3 The next-next-nearest-neighbour ($k=3$) walk

When $k=3$ the generating function of the Fourier unbounded propagator in Eq. (S9) can be written as

$$\begin{aligned}\widetilde{\widehat{P}}^{(3)}_{n_0}(\xi,z) &= \frac{e^{i\xi n_0}}{1-z\left\{1-q+\frac{q}{3}\Big[\cos(3k)+\cos(2k)+\cos(\xi)\Big]\right\}} \\ &= \frac{e^{i\xi n_0}}{1-z\left\{1-\frac{4q}{3}+\frac{4q}{3}\left[\cos^3(\xi)+\frac{1}{2}\cos^2(\xi)-\frac{1}{2}\cos(\xi)\right]\right\}} \\ &= \frac{3}{4zq}\frac{e^{i\xi n_0}}{1+\frac{3}{4q}\left[\frac{1}{z}-1\right]-\left[\cos^3(\xi)+\frac{1}{2}\cos^2(\xi)-\frac{1}{2}\cos(\xi)\right]} \\ &= \frac{3}{4zq}\left\{\frac{A_0}{\lambda_0-\cos(\zeta)}+\frac{B_+}{\lambda_+-\cos(\zeta)}+\frac{B_-}{\lambda_--\cos(\zeta)}\right\}e^{i\xi n_0},\end{aligned} \tag{S16}$$

where the λ's are the roots of the cubic equation in $\cos(\xi)$, namely

$$\begin{aligned}\lambda_0 &= -\frac{1}{6}-\frac{7\zeta^{-1}+\zeta}{6}, \\ \lambda_\pm &= -\frac{1}{6}+\frac{7\zeta^{-1}+\zeta}{12}\pm i\sqrt{3}\frac{7\zeta^{-1}-\zeta}{12},\end{aligned} \tag{S17}$$

with

$$\zeta = \left\{10-108\left[1+\frac{3}{4q}\left(\frac{1}{z}-1\right)\right]+\frac{9}{qz}\sqrt{3^4(1-z)^2+\frac{7^3}{3}q^2z^2+14^2qz(1-z)}\right\}^{1/3}. \tag{S18}$$

Given that $\lambda_0+\lambda_++\lambda_-=\lambda_+\lambda_-+\lambda_0(\lambda_++\lambda_-)=-\frac{1}{2}$ and $\lambda_0\lambda_+\lambda_-=1+\frac{3}{4q}\left(\frac{1}{z}-1\right)$ one can show that

$$\begin{aligned}A_0 &= \frac{1}{(\lambda_0-\lambda_+)(\lambda_0-\lambda_-)}=\frac{12\zeta^2}{\zeta^4+7\zeta^2+7^2}, \\ B_\pm &= \frac{1}{(\lambda_\pm-\lambda_0)(\lambda_\pm-\lambda_\mp)}=\frac{6\zeta^2}{\zeta^4+7\zeta^2+7^2}\left[-1\pm i\sqrt{3}\,\frac{\zeta^2+7}{\zeta^2-7}\right].\end{aligned} \tag{S19}$$

Using Eqs. (S16)-(S19) and noting that for $z\to 0$ we have $\zeta\simeq 7(qz/6)^{1/3}/3$, one finds that $\widetilde{\widehat{Q}}(\xi,0)=e^{i\xi n_0}$, and that $\widetilde{\widehat{Q}}(0,z)=A_0/(\lambda_0-1)+B_+/(\lambda_+-1)+B_-/(\lambda_--1)=(1-z)^{-1}$, indicating, respectively, that the occupation probability reduces to the initial condition at $t=0$ and that the propagator is normalised in space.

The propagator generating function in unbounded space, given by

$$\widetilde{P}^{(3)}_{n_0}(n,z)=\frac{3}{8\pi zq}\int_{-\pi}^{\pi}\mathrm{e}^{-i\xi(n-n_0)}\left\{\frac{A_0}{\lambda_0-\cos(\zeta)}+\frac{B_+}{\lambda_+-\cos(\zeta)}+\frac{B_-}{\lambda_--\cos(\zeta)}\right\}\mathrm{d}\xi, \tag{S20}$$

is used, analogously to the $k\in\{1,2\}$ cases, to derive via the method of images the bounded propagator

$$\begin{aligned}\widetilde{Q}^{(3)}_{n_0}(n,z) &= \frac{1}{N(1-z)}+\frac{3}{4zqN}\left\{A_0\sum_{k=1}^{N-1}\frac{\cos\left[\frac{2\pi k(n-n_0)}{N}\right]}{\lambda_0-\cos\left(\frac{2\pi k}{N}\right)}+B_+\sum_{k=1}^{N-1}\frac{\cos\left[\frac{2\pi k(n-n_0)}{N}\right]}{\lambda_+-\cos\left(\frac{2\pi k}{N}\right)}+B_-\sum_{k=1}^{N-1}\frac{\cos\left[\frac{2\pi k(n-n_0)}{N}\right]}{\lambda_--\cos\left(\frac{2\pi k}{N}\right)}\right\} \\ &= \frac{3}{4zq}\left\{A_0\frac{T_{N-|n-n_0|}(\lambda_0)+T_{|n-n_0|}(\lambda_0)}{(\lambda_0^2-1)\,U_{N-1}(\lambda_0)}+B_+\frac{T_{N-|n-n_0|}(\lambda_+)+T_{|n-n_0|}(\lambda_+)}{(\lambda_+^2-1)\,U_{N-1}(\lambda_+)}+B_-\frac{T_{N-|n-n_0|}(\lambda_-)+T_{|n-n_0|}(\lambda_-)}{(\lambda_-^2-1)\,U_{N-1}(\lambda_-)}\right\},\end{aligned} \tag{S21}$$

where, once again, identities linking finite trigonometric series to Chebyshev polynomials [3] have been used to obtain the last expression.

II MEAN FIRST-PASSAGE FOR THE K-NEIGHBOUR WALK ON THE RING LATTICE

We present here explicit expressions for the random walk mean first-passage time (MFPT) on the ring lattice. For the cases presented above, that is for $k \in \{1, 2, 3\}$, we show how the MFPT

$$\mathcal{F}^{(k)}_{n_0\to n} = \frac{k}{q}\sum_{\ell=1}^{N-1} \frac{\left[1-\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)\right]\sin\left(\frac{\ell\pi}{N}\right)}{k\sin\left(\frac{\ell\pi}{N}\right)-\sin\left(\frac{k\ell\pi}{N}\right)\cos\left(\frac{(k+1)\ell\pi}{N}\right)}, \tag{S22}$$

which we have slightly rewritten from Eq. (4) of the main text, i.e., here we consider the $q \in (0, 1]$ case, can be expressed in an alternative form, that is we are able to perform the finite summation. We also consider the extreme case when K covers the entire ring, that is when the walker may jump from one lattice site to any other lattice site.

II.A The nearest-neighbour ($k = 1$) walk

In this case Eq. (S22) reduces to

$$\mathcal{F}^{(1)}_{n_0\to n} = \frac{1}{q}\sum_{\ell=1}^{N-1} \frac{1-\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)}{1-\cos\left(\frac{2\ell\pi}{N}\right)}, \tag{S23}$$

which has been shown in ref. [3] to be equal to

$$\mathcal{F}^{(1)}_{n_0\to n} = \frac{1}{q}\Big(N-|n-n_0|\Big)|n-n_0|. \tag{S24}$$

II.B The next-nearest-neighbour ($k = 2$) walk

For this case, we exploit the explicit expression of the propagator derived in the last line of Eq. (S15), rather than Eq. (2) of the main text, and via the renewal relation [2, 5], $\widetilde{F}^{(2)}_{n_0}(n,z) = \frac{\widetilde{Q}^{(2)}_{n_0}(n,z)}{\widetilde{Q}^{(2)}_{n}(n,z)}$, we obtain the first-passage generating function

$$\begin{aligned}\widetilde{F}^{(2)}_{n_0}(n,z) = &\Bigg\{\Big[T_{N-|n-n_0|}(\beta_z^-)+T_{|n-n_0|}(\beta_z^-)\Big]\Big[(\beta_z^+)^2-1\Big]U_{N-1}(\beta_z^+)\\ &+(-1)^{n-n_0}\Big[T_{N-|n-n_0|}(\beta_z^+)+(-1)^N T_{|n-n_0|}(\beta_z^+)\Big]\Big[(\beta_z^-)^2-1\Big]U_{N-1}(\beta_z^-)\Bigg\}\\ &\times\Bigg\{\Big[T_N(\beta_z^-)+1\Big]\Big[(\beta_z^+)^2-1\Big]U_{N-1}(\beta_z^+)+\Big[T_N(\beta_z^+)+(-1)^N\Big]\Big[(\beta_z^-)^2-1\Big]U_{N-1}(\beta_z^-)\Bigg\}^{-1}.\end{aligned} \tag{S25}$$

Via $\mathcal{F}^{(2)}_{n_0\to n} = \frac{d}{dz}\widetilde{F}^{(2)}_{n_0}(n,z)\Big|_{z=1}$ we obatin the MFPT

$$\begin{aligned}\mathcal{F}^{(2)}_{n_0\to n} = &\frac{2}{5\,q}\Big(N-|n-n_0|\Big)|n-n_0| + \frac{4\sqrt{5}}{25\,q}N\Bigg(\frac{\cosh\left[N\ln\left(\frac{3+\sqrt{5}}{2}\right)\right]+(-1)^N}{\sinh\left[N\ln\left(\frac{3+\sqrt{5}}{2}\right)\right]}\\ &\times\left\{1-(-1)^{n-n_0}\cosh\left[|n-n_0|\ln\left(\frac{3+\sqrt{5}}{2}\right)\right]\right\}+(-1)^{n-n_0}\sinh\left[|n-n_0|\ln\left(\frac{3+\sqrt{5}}{2}\right)\right]\Bigg).\end{aligned} \tag{S26}$$

Given that the sum and subtraction of cosh and sinh terms grow very fast with n, it is numerically more stable to rewrite Eq. (S26) as

$$\begin{aligned}\mathcal{F}^{(2)}_{n_0\to n} = &\frac{2}{5\,q}\Big(N-|n-n_0|\Big)|n-n_0| + \frac{4\sqrt{5}}{25\,q}N\Bigg(1-(-1)^{n-n_0}\left(\frac{2}{3+\sqrt{5}}\right)^{|n-n_0|}\\ &+2\frac{(-1)^N+\left(\frac{2}{3+\sqrt{5}}\right)^N}{1-\left(\frac{2}{3+\sqrt{5}}\right)^N}\left\{\left(\frac{2}{3+\sqrt{5}}\right)^N-\frac{(-1)^{n-n_0}}{2}\left[\left(\frac{2}{3+\sqrt{5}}\right)^{N-|n-n_0|}+\left(\frac{2}{3+\sqrt{5}}\right)^{N+|n-n_0|}\right]\right\}\Bigg).\end{aligned} \tag{S27}$$

II.C The next-next-nearest-neighbour ($k = 3$) walk

To find the MFPT from the limit $z \to 1$ of $\frac{d}{dz}\left[\widetilde{Q}^{(3)}_{n_0}(n,z)/\widetilde{Q}^{(3)}_{n}(n,z)\right]$ in this case, it is more convenient to assume that z is a real variable and rewrite the coefficients A_0, $B_\pm$, λ_0, and $\lambda_\pm$. This can be done by defining

$$\alpha = 1 + \frac{3}{4q}\left[\frac{1}{z} - 1\right], \tag{S28}$$

and finding the roots of the depressed cubic polynomial

$$\alpha - \frac{5}{54} - y^3 + \frac{7}{12}y = 0, \tag{S29}$$

obtained by substituting $y = \cos(\xi) + 1/6$ in the denominator of the third line of Eq. (S16), that is in $\alpha - \cos^3(\xi) - \cos^2(\xi)/2 + \cos(\xi)/2 = 0$. The roots of Eq. (S29) can be expressed with hyperbolic functions and provide the alternative form of the elements in Eq. (S21) when z is real and positive:

$$\lambda_0 = -\frac{1}{6} + \frac{\sqrt{7}}{3}\cosh(\upsilon), \tag{S30}$$

$$\begin{aligned}
\lambda_\pm &= -\frac{1}{6} - \frac{\sqrt{7}}{6}\left[\cosh(\upsilon) \mp i\sqrt{3}\sinh(\upsilon)\right],\\
\upsilon &= \frac{1}{3}\text{arccosh}\left(\frac{108\,\alpha - 10}{7^{\frac{3}{2}}}\right),\\
A_0 &= \frac{12}{7}\frac{1}{1 + 2\cosh(2\upsilon)},\\
B_\pm &= -\frac{6}{7}\frac{1 \mp i\sqrt{3}\coth(\upsilon)}{1 + 2\cosh(2\upsilon)}.
\end{aligned} \tag{S31}$$

In the limit $z \to 1$, we have that $\upsilon \to \upsilon^* = 3^{-1}\text{arccosh}[2\sqrt{7}]$, which in turns give $\lambda_0 \to 1$ when $z \to 1$ since $\cosh(\text{arccosh}[2\sqrt{7}]/3) = \sqrt{7}/2$. We thus conveniently write the first-passage probability as

$$\widetilde{F}_{n_0}(n,z) = \frac{h(n,n_0,\upsilon) + (\lambda_0 - 1)U_{N-1}(\lambda_0)g(n,n_0,\upsilon)}{h(n,n,\upsilon) + (\lambda_0 - 1)U_{N-1}(\lambda_0)g(n,n,\upsilon)}, \tag{S32}$$

where

$$\begin{aligned}
h(n,n_0,\upsilon) &= A_0\frac{T_{N-|n-n_0|}(\lambda_0) + T_{|n-n_0|}(\lambda_0)}{(\lambda_0 + 1)},\\
g(n,n_0,\upsilon) &= B_+\frac{T_{N-|n-n_0|}(\lambda_+) + T_{|n-n_0|}(\lambda_+)}{\left(\lambda_+^2 - 1\right)U_{N-1}(\lambda_+)} + B_-\frac{T_{N-|n-n_0|}(\lambda_-) + T_{|n-n_0|}(\lambda_-)}{\left(\lambda_-^2 - 1\right)U_{N-1}(\lambda_-)}.
\end{aligned} \tag{S33}$$

The terms that survive in the evaluation of the mean first-passage time $\mathcal{F}^{(3)}_{n_0\to n}$ gives

$$\begin{aligned}
\mathcal{F}^{(3)}_{n_0\to n} = \left.\frac{d\upsilon}{dz}\right|_{z=1} &\left\{\left[\frac{d}{d\upsilon}h(n,n_0,\upsilon)\right]h(n,n,\upsilon) - \left[\frac{d}{d\upsilon}h(n,n,\upsilon)\right]h(n,n_0,\upsilon)\right.\\
&\left. + \left(\frac{d}{d\upsilon}\left[(\lambda_0 - 1)U_{N-1}(\lambda_0)\right]\right)\left[g(n,n_0,\upsilon)h(n,n,\upsilon) - g(n,n,\upsilon)h(n,n_0,\upsilon)\right]\right\}\left\{h(n,n,\upsilon)\right\}^{-2}\Bigg|_{\upsilon=\upsilon^*}
\end{aligned} \tag{S34}$$

Quantities relevant to evaluate the above MFPT are the following

$$\begin{aligned}
&\left.\frac{dv}{dz}\right|_{z=1} = -\frac{1}{q}\left(\frac{3}{7}\right)^{3/2} \\
&h\left(n, n_0, v^*\right) = \frac{2}{7} \\
&\left.\frac{d}{dv} h\left(n, n_0, v\right)\right|_{v=v^*} = \frac{1}{2\sqrt{21}}\left[5 - N^2 - 2(N - |n - n_0|)|n - n_0|\right] \\
&\left.\frac{d}{dv}\left[(\lambda_0 - 1)U_{N-1}(\lambda_0)\right|_{v=v^*} = \frac{\sqrt{7}}{3} N \sinh\left(v^*\right) = \sqrt{\frac{7}{3}}\frac{N}{2} \\
&g\left(n, n_0, v^*\right) = \frac{\sqrt{6}}{7}\frac{\sqrt{1+\frac{i}{3\sqrt{7}}}\Big\{\cosh[(N - |n - n_0|)\omega] + \cosh[(n - n_0)\omega]\Big\}}{\sinh\left(N\omega\right)} + \text{c.c.}
\end{aligned} \tag{S35}$$

where $\omega = \operatorname{arcosh}\left(-\frac{3}{4} - i\frac{\sqrt{7}}{4}\right)$. This allows to write

$$\mathcal{F}^{(3)}_{n_0\to n} = \frac{3}{14q}(N - |n - n_0|)|n - n_0| + \frac{3N}{4q}\Big[g(n, n, v^*) - g(n, n_0, v^*)\Big]. \tag{S36}$$

To extract directly the real part of Eq. (S36), one converts to polar coordinates in the complex domain and rewrite

$$\begin{aligned}
&\sqrt{1 \pm \frac{i}{3\sqrt{7}}} = \frac{1}{\sqrt{2}(63)^{1/4}}\left[\sqrt{8+3\sqrt{7}} \pm i\sqrt{8-3\sqrt{7}}\right], \\
&\operatorname{arcosh}\left(-\frac{3}{4} \pm i\frac{\sqrt{7}}{4}\right) = \frac{1}{2}\ln\left(\frac{2+\sqrt{7}+\sqrt{7+4\sqrt{7}}}{2}\right) \pm i\left[\pi - \arctan\left(\frac{\sqrt{7+4\sqrt{7}}}{3}\right)\right]
\end{aligned} \tag{S37}$$

and using the following definition

$$\begin{aligned}
f_1(m) &= \cosh(m\rho)\cos(m\beta), \\
f_2(m) &= \cosh(m\rho)\sin(m\beta), \\
f_3(m) &= \sinh(m\rho)\cos(m\beta), \\
f_4(m) &= \sinh(m\rho)\sin(m\beta),
\end{aligned} \tag{S38}$$

with $\rho = \frac{1}{2}\ln\left(\frac{2+\sqrt{7}+\sqrt{7+4\sqrt{7}}}{2}\right)$, $\beta = \arctan\left(\frac{\sqrt{7+4\sqrt{7}}}{3}\right)$, and $m \geq 0$ an integer. The ensuing identity

$$\begin{aligned}
&\left(\sqrt{8+3\sqrt{7}} + i\sqrt{8-3\sqrt{7}}\right)\cosh\left\{m\operatorname{arccosh}\left[\frac{-3+i\sqrt{7}}{4}\right]\right\}\sinh\left\{N\operatorname{arccosh}\left[\frac{-3-i\sqrt{7}}{3}\right]\right\} \\
&+\left(\sqrt{8+3\sqrt{7}} - i\sqrt{8-3\sqrt{7}}\right)\cosh\left\{m\operatorname{arccosh}\left[\frac{-3-i\sqrt{7}}{4}\right]\right\}\sinh\left\{N\operatorname{arccosh}\left[\frac{-3+i\sqrt{7}}{3}\right]\right\} \\
&= 2(-1)^{N+m}\left\{\sqrt{8+3\sqrt{7}}\Big[f_1(m)f_3(N) + f_2(N)f_4(m)\Big] - \sqrt{8-3\sqrt{7}}\Big[f_1(m)f_2(N) - f_3(N)f_4(m)\Big]\right\},
\end{aligned} \tag{S39}$$

is employed to obtain finally

$$\begin{aligned}
\mathcal{F}^{(3)}_{n_0\to n} &= \frac{1}{q}\frac{3}{14}\Big(N - |n - n_0|\Big)|n - n_0| \\
&+ \frac{N}{2q}\frac{3}{7^{5/4}}\frac{\mathcal{G}(N) + (-1)^N\mathcal{G}(0) - (-1)^{|n-n_0|}\Big[\mathcal{G}(N - |n - n_0|) + (-1)^N\mathcal{G}(|n - n_0)|)\Big]}{1 + \sigma^{4N} - 2\sigma^{2N}\cos\left[2N\ln(\sigma)\right]},
\end{aligned} \tag{S40}$$

where

$$\begin{aligned}
\mathcal{G}(m) &= \sigma^{N-m}\sqrt{8+3\sqrt{7}}\Big\{\Big(\sigma^{2m} - \sigma^{2N}\Big)\cos\Big[(N+m)\beta\Big] + \Big(1 - \sigma^{2(m+N)}\Big)\cos\Big[(N-m)\beta\Big]\Big\} \\
&- \sigma^{N-m}\sqrt{8-3\sqrt{7}}\Big\{\Big(1 + \sigma^{2(N+m)}\Big)\sin\Big[(N+m)\beta\Big] + \Big(\sigma^{2N} + \sigma^{2m}\Big)\sin\Big[(N-m)\beta\Big]\Big\}
\end{aligned} \tag{S41}$$

with $\sigma = \sqrt{\frac{2}{2+\sqrt{7}+\sqrt{7+4\sqrt{7}}}}$.

II.D The lattice walk MFPT on the all-to-all graph for odd N

A simple way to derive the expression for this case from Eq. (S22) is to consider N odd and to take $k = (N-1)/2$. One then obtains

$$\mathcal{F}_{n_0\to n}^{\left(\frac{N-1}{2}\right)} = \frac{1}{q}\sum_{\ell=1}^{N-1} \frac{\left[1-\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)\right]}{1-\frac{2}{N-1}\frac{\sin\left(\frac{(N-1)\ell\pi}{2N}\right)\cos\left(\frac{(N+1)\ell\pi}{2N}\right)}{\sin\left(\frac{\ell\pi}{N}\right)}}. \tag{S42}$$

Exploiting the relation $\sin(A)\cos(B) = \frac{\sin(A+B)+\sin(A-B)}{2}$ it is straightforward to simplify Eq. (S42) to

$$\mathcal{F}_{n_0\to n}^{\left(\frac{N-1}{2}\right)} = \frac{N-1}{Nq}\sum_{\ell=1}^{N-1}\left[1-\cos\left(\frac{2\pi\ell(n-n_0)}{N}\right)\right], \tag{S43}$$

where the finite summation is now trivial to evaluate leading to

$$\mathcal{F}_{n_0\to n}^{\left(\frac{N-1}{2}\right)} = \frac{1}{q}(1-\delta_{n,n_0})(N-1) \tag{S44}$$

given in the main text. To consider also the case when N is even, we have first derived the dynamics of the occupation probability for any N and then extracted the MFPT as we show below.

III OCCUPATION PROBABILITY DYNAMICS FOR THE LATTICE WALK ON THE ALL-TO-ALL GRAPH FOR ODD AND EVEN N

We start by identifying the lattice walk transition matrix $\mathbb{S}$ for the all-to-all network. Its elements are $\mathbb{S}_{i,i} = 1-q$, which represents the probability to stay at a node, while all the off-diagonal elements, which describe the probability to move to any of the other nodes are $\mathbb{S}_{i,j} = \frac{q}{N-1}$ for $i \neq j$, dictated by conservation of probability. Such a matrix can be conveniently expressed as

$$\mathbb{S} = \frac{q}{N-1}\mathbb{T} \tag{S45}$$

with $\mathbb{T}_{i,i} = (N-1)(1-q)/q$ and $\mathbb{T}_{i,j} = 1$ for $i \neq j$. The Master equation in vectorial form is given by

$$\boldsymbol{\phi}(t+1) = \mathbb{S}\cdot\boldsymbol{\phi}(t), \tag{S46}$$

where we use $\phi(n,t)$ to denote the lattice walk occupation probability when constrained to the all-to-all graph. Equation S46 is solved through eigendecomposition as

$$\boldsymbol{\phi}(t) = \mathbb{S}^t\cdot\boldsymbol{\phi}(0) = \left(\frac{q}{N-1}\right)^t\mathbb{T}^t\cdot\boldsymbol{\phi}(0) = \left(\frac{q}{N-1}\right)^t\left(\boldsymbol{A}\cdot\boldsymbol{\Lambda}\cdot\boldsymbol{A}^{-1}\right)^t\cdot\boldsymbol{\phi}(0) = \left(\frac{q}{N-1}\right)^t\boldsymbol{A}\cdot\boldsymbol{\Lambda}^t\cdot\boldsymbol{A}^{-1}\cdot\boldsymbol{\phi}(0), \tag{S47}$$

where $\boldsymbol{\Lambda}$ is the diagonal matrix of eigenvalues of $\mathbb{T}$, namely $\boldsymbol{\Lambda}_{ij} = 0$ for $i \neq j$, $\boldsymbol{\Lambda}_{11} = (N-1)/q$ and $\boldsymbol{\Lambda}_{ii} = [N(1-q)-1]/q$ for $i = 2, ..., N$. $\boldsymbol{A}$ is the corresponding matrix of normalised right eigenvectors $\boldsymbol{A}$, whose elements are $\boldsymbol{A}_{i,1} = -\frac{1}{\sqrt{N}}$ for $i = 2, ..., N$, $\boldsymbol{A}_{1,j} = \frac{1}{\sqrt{N}}$ for $j = 1, ..., N$, $\boldsymbol{A}_{i+1,N+1-i} = \frac{1}{\sqrt{N}}$ for $i = 1, ..., N-1$ and 0 elsewhere. For the inverse we have instead $\boldsymbol{A}^{-1}_{1,j} = \frac{1}{\sqrt{N}}$, $\boldsymbol{A}^{-1}_{i+1,N+1-i} = \frac{N-1}{\sqrt{N}}$ for $i = 1, ..., N-1$ and $-\frac{1}{\sqrt{N}}$ elsewhere.

After a bit of algebra and defining $\boldsymbol{A}^{-1}\cdot\boldsymbol{\Lambda}^t\cdot\boldsymbol{A} = \mathbb{K}$ one gets

$$\mathbb{K}_{i,i} = \frac{1}{N}\left[\left(\frac{N-1}{q}\right)^t + (N-1)\left(\frac{(N-1)(1-q)}{q}-1\right)^t\right], \tag{S48}$$

and for $i \neq j$

$$\mathbb{K}_{i,j} = \frac{1}{N}\left[\left(\frac{N-1}{q}\right)^t - \left(\frac{(N-1)(1-q)}{q}-1\right)^t\right]. \tag{S49}$$

Inserting (S48) and (S49) in (S47) and considering a starting condition with a walker localised at $n = n_0$, that is $\phi(n,0) = \delta_{n,n_0}$, one obtains the simple result

$$\phi_{n_0}(n,t) = \frac{1}{N}\left[1 - \left(1 - q - \frac{q}{N-1}\right)^t\right] + \left(1 - q - \frac{q}{N-1}\right)^t \delta_{n,n_0}, \tag{S50}$$

which is properly normalised over all the nodes of the network and satisfy the initial condition. Note that for the case $q = (N-1)/N$, which corresponds to the jump probabilities being all equal including the self loops, Eq. (S50) reduces to the trivial dynamics

$$\phi_{n_0}(n,t) = \frac{1}{N}\left[1 - \delta_{t,0}\right] + \delta_{t,0}\delta_{n,n_0}, \tag{S51}$$

III.A First-passage probability and MFPT

It is a trivial exercise now to take the generating function of (S50), namely

$$\begin{aligned}\widetilde{\phi}_{n_0}(n,z) &= \frac{1}{N}\left[\frac{1}{1-z} - \frac{1}{1 - z\left(1 - q - \frac{q}{N-1}\right)}\right] + \frac{1}{1 - z\left(1 - q - \frac{q}{N-1}\right)}\delta_{n,n_0} \\ &= \frac{1}{N}\left[\frac{1}{1-z} + \frac{N\delta_{n,n_0} - 1}{1 - z\left(1 - q - \frac{q}{N-1}\right)}\right]\end{aligned} \tag{S52}$$

and write for $n \neq n_0$ the first-passage probability generating function as

$$\widetilde{F}^{(\mathrm{all})}_{n_0}(n,z) = \frac{\widetilde{\phi}_{n_0}(n,z)}{\widetilde{\phi}_n(n,z)} = \frac{q}{N-1}\frac{z}{1 - z\left(1 - \frac{q}{N-1}\right)}. \tag{S53}$$

Inverting Eq. (S53) to time gives

$$F^{(\mathrm{all})}_{n_0}(n,t) = \frac{q}{N-1}\left(1 - \frac{q}{N-1}\right)^{t-1}, \qquad t \geq 1, \tag{S54}$$

and $F^{(\mathrm{all})}_{n_0}(n,0) = 0$, which, as expected, is independent of n. For the special case where all jump probabilities are equal, i.e. $q = (N-1)/N$, Eq. (S54) reduces to

$$F^{(\mathrm{all})}_{n_0}(n,t) = \frac{(1 - 1/N)^{t-1}}{N}, \qquad t \geq 1, \tag{S55}$$

while the case $q = 1$ gives

$$F^{(\mathrm{all})}_{n_0}(n,t) = \frac{1}{N-1}\left(\frac{N-2}{N-1}\right)^{t-1}, \qquad t \geq 1, \tag{S56}$$

which was reported (without derivation) in ref. [6] as $\frac{1}{N-1}\left(\frac{N-2}{N-1}\right)^t$.

For the MFPT it is straightforward to obtain from (S54)

$$\mathcal{F}^{(\mathrm{all})}_{n_0 \to n} = \frac{N-1}{q}, \tag{S57}$$

valid for $n \neq n_0$.

IV OCCUPATION PROBABILITY ON THE SMALL-WORLD NETWORK

In Eq. (6) of the main text we have presented the Master equation governing the dynamics of $S(n,t)$ on SWN, where the network is constructed by rewiring links of K-neighbour ring lattice. The rewiring makes the ring lattice spatially

disordered, with spatial heterogeneities of the inert type (probability preserving). The rewiring are parameterised via a set of four-tuples $(u, v, \eta_{v,u}, \eta_{u,v})$ with u and v denoting a pair of sites where the outgoing connections between them are modified by $\eta_{v,u}$, $\eta_{u,v}$, respectively. With defective sites being of the inert type, the general formalism of ref. [7] that allows to describe the dynamics in disordered lattices can be employed. Through that formalism we derive analytically the propagator of the Master equation as the generating function

$$\widetilde{S}_{n_0}(n,z) = \widetilde{Q}^{(k)}_{n_0}(n,z) - 1 + \frac{\det[\mathbb{H}(n,n_0,z)]}{\det[\mathbb{H}(z)]}, \tag{S58}$$

where the matrices of size $M \times M$ are as follows

$$\mathbb{H}(z)_{i,j} = \eta_{u_i,v_i}\widetilde{Q}^{(k)}_{\langle u_j - v_j\rangle}(u_i,z) - \eta_{v_i,u_i}\widetilde{Q}^{(k)}_{\langle u_j - v_j\rangle}(v_i,z) - \frac{\delta_{i,j}}{z}, \tag{S59}$$

and

$$\mathbb{H}(n,n_0,z)_{i,j} = \mathbb{H}(z)_{i,j} - \widetilde{Q}^{(k)}_{\langle u_j - v_j\rangle}(n,z)\left[\eta_{u_i,v_i}\widetilde{Q}^{(k)}_{n_0}(u_i,z) - \eta_{v_i,u_i}\widetilde{Q}^{(k)}_{n_0}(v_i,z)\right]. \tag{S60}$$

Note that the difference of the (defect-free) occupation probability evaluated at sites where the transition probabilities have been modified enter the evaluation of $\widetilde{S}_{n_0}(n,z)$, and it is apparent in terms represented through the notation $f_{\langle u-v\rangle}(\cdot) = f_u(\cdot) - f_v(\cdot)$.

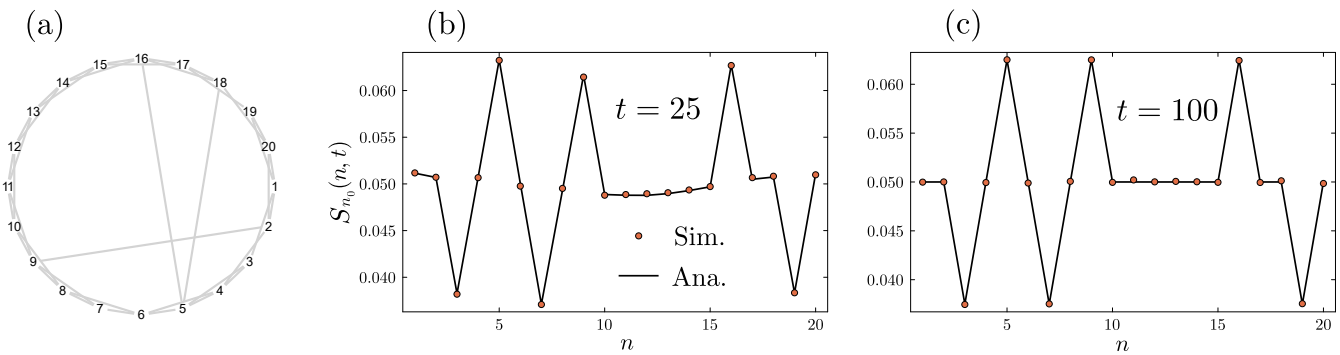

FIG. S1. A comparison between the theoretically predicted occupation probability, extracted from Eq. S58, and the average of 10^6 simulations at (b) $t = 25$ and (c) $t = 100$ in a specific SWN realisation generated from a $K = 4$ ring lattice with $p = 0.15$ and $N = 20$ depicted in (a). The initial condition is $n_0 = 1$.

In Fig. S1 we compare against stochastic simulations the dynamics of $S_{n_0}(n,t)$ extracted by inverting numerically in time through a straightforward one-dimensional trapezoidal integration [8] of Eq. (S58). The plot shows that the occupation probability in the SWN is qualitatively very different from the broadening of a single peaked function centred on n_0 that one would expect at short times on the ring lattice. $S_{n_0}(n,t)$ instead displays a complex dependence on the degree of the node and the initial condition, with some nodes with probability values already approaching the steady state after only 25 steps, while some nodes that are near or in correspondence of the nodes with long range connections remain farther from it. Note that for the steady state we have [9] $S_{n_0}(n, t \to \infty) = \chi_n/\mathcal{E}$, with χ_n the coordination number of node n and $\mathcal{E}$ the number of edges in the network.

V MEAN FIRST-PASSAGE AND FIRST-ABSORPTION TIME ON THE SMALL-WORLD NETWORK

The MFPT on the network can also be derived algebraically from the occupation probability in Eq. (S58). The implementation of the general MFPT formalism from ref. [7]) in the SWN context gives

$$\mathcal{F}^{(k)}_{n_0\to n} = \frac{\mathcal{F}^{(k)}_{n_0\to n}\det\left[\mathbb{L} - \left(1/\mathcal{F}^{(k)}_{n_0\to n}\right)\mathbb{M}\right]}{\det\left[\mathbb{L} - \mathbb{O}\right]}, \tag{S61}$$

where the matrices of size $M \times M$ are

$$\mathbb{L}_{i,j} = \frac{\eta_{v_i,u_i}}{\mathcal{R}_{u_i}^{(k)}} \mathcal{F}_{\langle u_j - v_j \rangle \to u_i}^{(k)} - \frac{\eta_{u_i,v_i}}{\mathcal{R}_{v_i}^{(k)}} \mathcal{F}_{\langle u_j - v_j \rangle \to v_i}^{(k)} + \delta_{i,j}, \tag{S62}$$

$$\mathbb{M}_{i,j} = \left(\frac{\eta_{v_i,u_i}}{\mathcal{R}_{u_i}^{(k)}} \mathcal{F}_{\langle n_0 - n \rangle \to u_i}^{(k)} - \frac{\eta_{u_i,v_i}}{\mathcal{R}_{v_i}^{(k)}} \mathcal{F}_{\langle n_0 - n \rangle \to v_i}^{(k)} \right) \mathcal{F}_{\langle u_j - v_j \rangle \to n}^{(k)}, \tag{S63}$$

$$\mathbb{O}_{i,j} = \left(\frac{\eta_{v_i,u_i}}{\mathcal{R}_{u_i}^{(k)}} - \frac{\eta_{u_i,v_i}}{\mathcal{R}_{v_i}^{(k)}} \right) \mathcal{F}_{\langle u_j - v_j \rangle \to n}^{(k)}. \tag{S64}$$

Note that the MFPT $\mathcal{F}_{n_0 \to n}^{(k)}$ depends on the disorder parameters and the MFPT, but also on the mean return time (MRT), in the ring lattice.

When the target is partially absorbing ($0 < \rho < 1$), the above MFPT expression is used to calculate the mean-first absorption time (MFAT) from n_0 to n, whose general expression is given by (see e.g. [10, 11])

$$\mathcal{A}_{n_0 \to n}(\rho) = \mathcal{F}_{n_0 \to n}^{(k)} + \frac{1-\rho}{\rho} \mathcal{R}_n, \tag{S65}$$

with the MRT being $\mathcal{R}_n = \mathcal{E}/\chi_n$.

VI FIRST-PASSAGE PROBABILITY AND MEAN FIRST-ABSORPTION TIME FOR THE DEFECTIVE RING LATTICE

As described in the main text, we consider a ring lattice with each lattice site having K links to the nearest neighbours and may include one self loop into itself. We modify such lattice to create a special network by placing one short-cut between nodes $n+1$ and n_0+5, which requires the use of $M = K+1$ defects in the the exact formalism described in Sec. IV. Using Eq. (S58) for $\widetilde{S}_{n_0}(n,z)$ we are able to study the dynamics of first-absorption from site n_0 to site n in this special network via

$$\widetilde{A}_{n_0}(n,z) = \frac{\rho\, \widetilde{S}_{n_0}(n,z)}{1-\rho+\rho\, \widetilde{S}_n(n,z)}. \tag{S66}$$

We numerical invert to time [8] Eq. (S66) and plot the first-absorption dynamics for different choice of N and ρ in

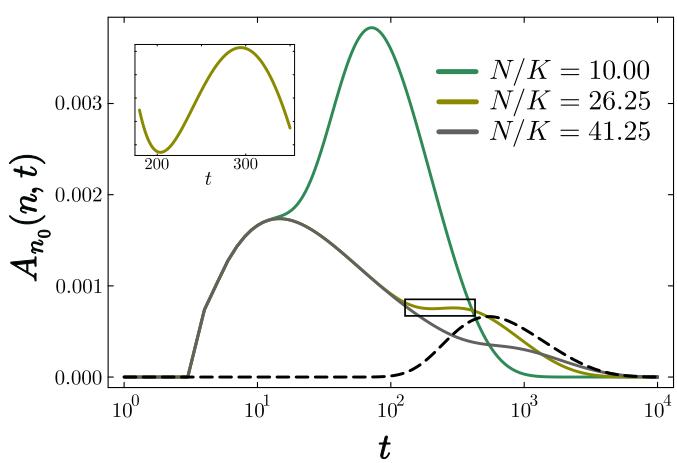

FIG. S2. Temporal dependence of the first-absorption probability from $n_0 = 1$ to the partially absorbing site ($\rho = 0.85$) at $n = N/2$ for the above described network with $K = 8$ and different values of N. The dashed curve represents the case without any rewiring of the original ring lattice. The inset is a blow up of the region $180 < t < 340$ for the $N/K = 26.25$ case.

Fig. S2. For one parameter selection, namely $N/K = 10.00$, we have chosen N small enough so that it falls just below the regime of bi-modality (see Fig. 3(b) in the main text). One may notice that a change of flex after $t = 10$ steps, which is indicative of the fact the network is so small that the trajectories that eventually get absorbed at the target site and take the shortcut and those that move around the ring are of comparable timescales, which gives rise to the

pronounced mode. For a bigger network, namely $N = 210$ ($N/K = 26.25$), we enter the bi-modal regime, and the above timescales can now be differentiated: the network is large enough for the first mode to appear in correspondence of the change of flex of the network with $N = 80$. The third choice of network size, namely $N = 330$ ($N/K = 41.25$) is above the bi-modal region and one observe that the mode due to the direct trajetcories is still present, but the second one has disappeared. (For other choices of N and ρ we have been presented plots in Fig. 3(c) of the main text.)

The analytic formula in Eq (S65) allows us to study the mean first-absorption time as a function of the network size and compare it to the value of the first and second mode of the first-absorption probability. While in the main text we have shown such comparison in Fig 3(d) when $\rho = 1$, in Fig. S3 we present the case $\rho = 0.5$. A comparison to the corresponding plot in the main text, shows a rightward shift and a narrowing of the region of bimodality, which is consistent with what observed in Fig. 3(b) of the main text.

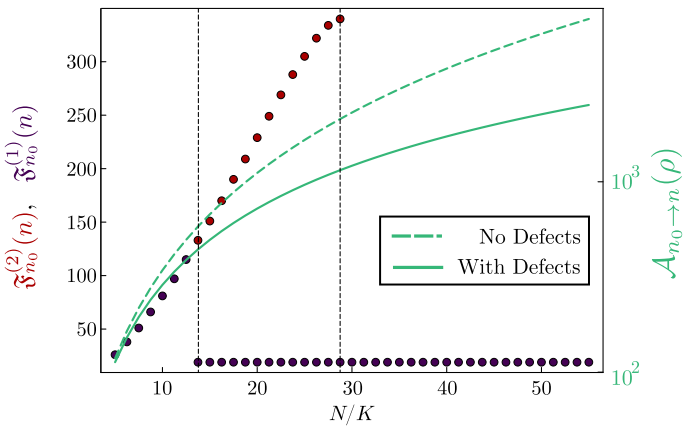

FIG. S3. Dependence of the first and second mode of the first-absorption probability, respectively, $\mathfrak{F}^{(1)}_{n_0}(n)$ and $\mathfrak{F}^{(2)}_{n_0}(n)$, and the MFAT in Eq. (S65) for the small world network selected in the main text in Fig. 3 when $\rho = 0.5$ and $K = 8$ as a function of the network size N. The choice of n_0 and n are described in Fig. S2. The specific choice of the network and the target location n gives $\mathcal{R}_n = N$. The region in between the two vertical dashed lines corresponds to when both modes are present. The green solid represents the MFAT in the network in the absence of the short-cut, that is the MFAT in the ring lattice.

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